

HoneCollege

Student Name: _____

Date: _____

Year 10 Mathematics

Topic: Non-linear relationships, functions and their graphs

Theory Booklet #1

Learning Milestones

- Understand the shape and features of parabolas
- Sketch parabolas using transformations
- Sketch parabolas using factorisation
- Sketch parabolas using completing the square
- Interpret graphs using quadratic formula and discriminant

Exploring Parabolas

A parabola is the graph of a quadratic function:

$$y = ax^2 + bx + c$$

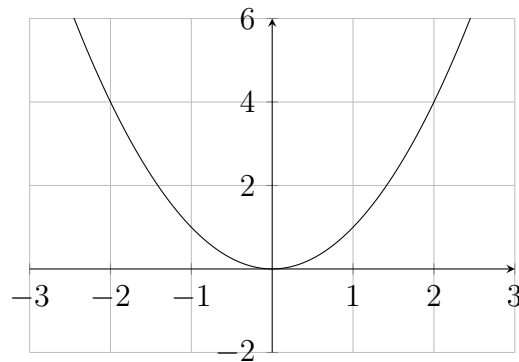
Key features:

- Vertex (turning point)
- Axis of symmetry
- x-intercepts
- y-intercept

Shape depends on a :

- $a > 0$ opens upwards
- $a < 0$ opens downwards

Diagram: Basic Parabola



Example: Identify key features of $y = x^2$

Basic Parabolic Shifts

Starting from:

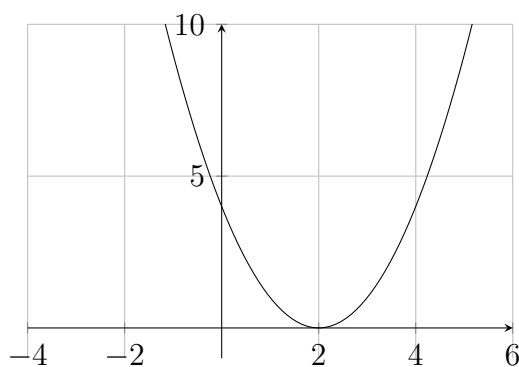
$$y = x^2$$

We apply shifts to create new parabolas.

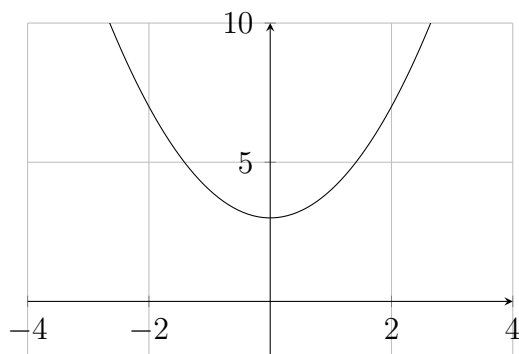
Key idea:

- $(x - h)$ shifts right
- $(x + h)$ shifts left
- $+k$ shifts up
- $-k$ shifts down

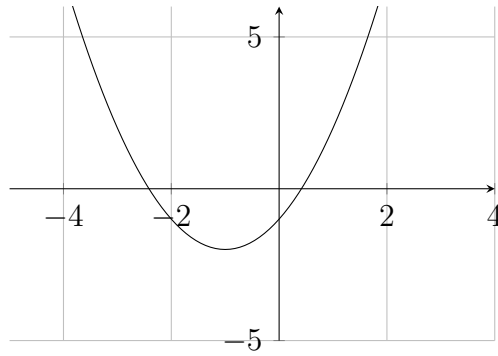
1. Right shift: $y = (x - 2)^2$



2. Up shift: $y = x^2 + 3$



3. Left and down shift: $y = (x + 1)^2 - 2$



1. Sketch $y = (x - 3)^2$

2. Sketch $y = x^2 - 4$

Sketching Using Factorisation

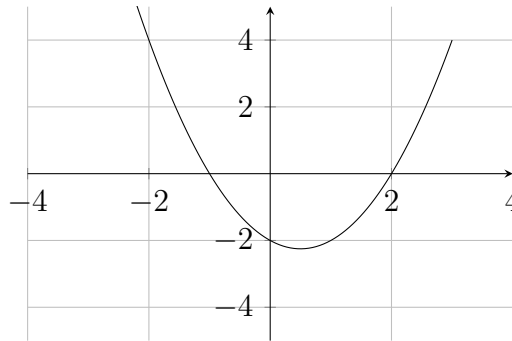
If:

$$y = (x - p)(x - q)$$

Then:

- x-intercepts are p and q
- axis of symmetry is midpoint of p and q

Example: $y = (x - 2)(x + 1)$



1. Sketch $y = (x + 3)(x - 1)$

2. Sketch $y = (x + 2)(x + 5)$

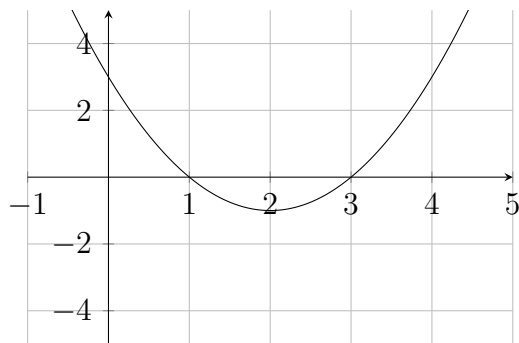
Completing the Square

Vertex form:

$$y = a(x - h)^2 + k$$

Vertex is (h, k) .

Example: $y = x^2 - 4x + 3$



1. Rewrite and sketch $y = x^2 - 6x + 5$

2. Rewrite and sketch $y = x^2 - 2x - 8$

Quadratic Formula and Discriminant

Quadratic formula:

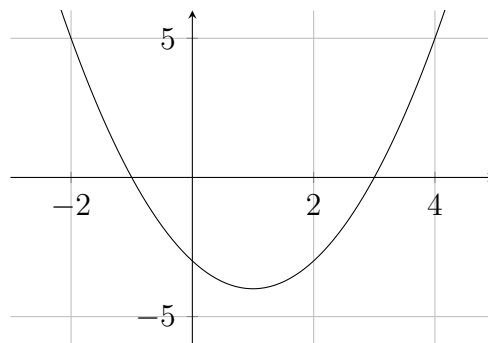
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Discriminant:

$$D = b^2 - 4ac$$

- $D > 0$ two roots
- $D = 0$ one root
- $D < 0$ no real roots

Example: $y = x^2 - 2x - 3$



1. Sketch the graph of $y = x^2 - 4x - 5$ and hence determine:
 - the x-intercepts
 - whether the graph has a maximum or minimum point

Graphing Practice — Mixed Quadratics

Sketch each parabola carefully. Draw axes, and mark key features.

1. $y = x^2$

4. $y = x^2 - 4x + 3$

2. $y = (x - 2)^2$

5. $y = (x - 1)^2 + 2$

3. $y = (x + 1)(x - 3)$

6. $y = x^2 + 3x$

7. $y = x^2 - 2x - 3$

10. $y = x^2 - 9$

8. $y = x^2 + 2x + 1$

11. $y = (x + 2)(x - 4)$

9. $y = (x - 3)^2 - 2$

12. $y = x^2 - x - 6$